

WASHINGTON UNIVERSITY IN ST. LOUIS
OLIN BUSINESS SCHOOL
DAT 5564: Database Design and SQL
Fall 2025
Homework #1: E/R Diagrams

Instructions

1. This is an individual assignment. You may not discuss your approach to solving these questions with anyone other than the instructor or TA.
2. Please include only your Student ID on the PDF submission.
3. The only allowed material is:
 - a. Class notes
 - b. Content posted on Canvas
 - c. Textbook
4. You are not permitted to use other online resources
5. Due online, before the next lab.
6. There will be TA office hours. See the schedule on Canvas.

Assignment

In addition to the questions on Canvas, please upload your diagram along with the answers to these questions in a PDF file. When preparing a diagram, please include a brief explanation of your logic. Clarify any assumptions you make. Handwritten solutions are fine.

Background

STM is a local movie theater house in St. Louis. The theater has several screens in the theater. It schedules movies for each screen for the following week.

The movies that are scheduled include new releases from the large movie studios, such as Disney and Warner Brothers. They also show classic movies and movies that target niche audiences. Occasionally, they show international movies that may have only limited interest in the St. Louis market. As another example, on weekends they often have children's movies during the day

The multiple screens in the theater have different capacities. This allows scheduling movies with smaller audiences in smaller theater spaces. As an example, a children's movie on the weekend would be scheduled in a large theater space, while the same movie on a weekday evening would be scheduled in a small theater.

In each day, there are multiple time slots. The time slots are usually spaced out every 3 hours. This allows showing the movie in its entirety, with sufficient commercials, and cleaning the theater once guests have left. For example, in each day, each theater space may have movies scheduled at 10 am, 1 pm, 4 pm, etc.

Another aspect of the theater spaces is that theaters vary on several dimensions. Some theaters have luxury, reclining seats or heated seats. Others have full food service available.

Database

Develop the E/R diagram for a database for STM, the local movie theater. For now, we'll focus on a few aspects of the theater spaces, movies, and movie scheduling. Later parts of this assignment will expand on the database. Not everything described above is covered here.

For movies, the database should include information on each movie. This includes the name of the movie, length, genre, rating, country of origin, and 2 other attributes.

In terms of the screens in the theater spaces, we want to know the number of seats in the theater, the number of handicap seats in the theater, the number of rows of seats, and whether the seats are luxury, reclining seats.

Some of the screens also offer a full menu. These are called “dining screens”. In this type of theater there is a full menu offered. Customers can order food from their seat. One of the theater employees will bring the food to the customer's seat. Some of the food items include: hot-dogs, popcorn, nachos, and drinks. For the dining screen, we only want to know the name of the manager, and whether it is open during the day.

Movies are scheduled for the different theaters. They are scheduled for all 7 days of the week. When a movie is scheduled in a theater space, it takes up that time slot for the 3 hour duration. Each movie may be shown more than once during the week. However, in each time slot, in each theater, only one movie can be shown.

As mentioned above, a children's movie may be scheduled for a daytime showing on the weekend, but for an evening showing during the week. If it is very popular, it may be shown on two screens at the same time on the same weekend day.

Part 1: [60 points]

Design the corresponding E/ER diagram. Remember throughout to include arrows representing cardinality, where relevant. Underline key attributes.

- i. Also, convert the diagram to relations. In other words, show the tables that would be used in the database. You do not need to include data, only the structure of the tables.
- ii. Suggest 3 queries that you could ask of this database.
- iii. Suggest 1 query that requires additional attributes. Which attributes are needed to answer this additional query?

Question 2: [20 points]

Modify the database to include more information about the dining screens. For each dining screen, we would like the list of all menu items offered. We don't need to keep prices or other data, only the menu items that are offered

Question 3: [20 points]

Modify the database to include more information about each movie. For each movie, we would like to know the names of the actors that appear in that movie. The information stored about actors is only about their role in that movie. (*Hint*: that's important for how you add actor information to the E/R diagram.) For the actor information, we are only interested in the actor's name, minutes in this movie, and salary in this movie.

Now include cardinality for the relations that you added.